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CSA1717 – ARTIFICIAL INTELLIGENCE

CO1 – LIST LAB PROGRAMS PYTHON

EXP 4: Crypt-Arithmetic Problem

PROGRAM:

practical lab 4.py - C:/Users/yashm/OneDrive/Desktop/PYTHON AI/practical lab 4.py (3.14.5)

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```
from itertools import permutations

letters = "SENDMORY"
digits = "0123456789"

for p in permutations(digits, len(letters)):
    d = dict(zip(letters, p))

    if d["S"] == "0" or d["M"] == "0":
        continue

    send = int(d["S"] + d["E"] + d["N"] + d["D"])
    more = int(d["M"] + d["O"] + d["R"] + d["E"])
    money = int(
        d["M"] + d["O"] + d["N"] + d["E"] + d["Y"]
    )

    if send + more == money:
        print("SEND =", send)
        print("MORE =", more)
        print("MONEY =", money)
        break
```

OUTPUT:

IDLE Shell 3.14.5

File Edit Shell Debug Options Window Help

```
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/yashm/OneDrive/Desktop/PYTHON AI/practical lab 4.py ====
SEND = 9567
MORE = 1085
MONEY = 10652
>>>
```